

Digital Design and Computer Organisation Laboratory

UE23CS251A

3rd Semester, Academic Year 2024-25

Date:17/10/2024

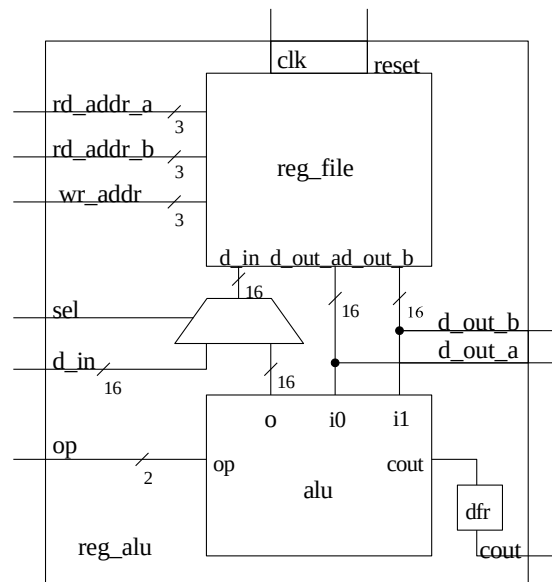
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Week# 6

Program Number: 1

Register File (of eight 16-bit registers)

- Verilog Code Screenshot
- Verilog VVP Output Screenshot
- ShotGTKWAVE Screenshot



reg_alu.v:-

```
reg_alu.v U X  tb_reg_alu.v U

week6 > reg_alu.v
1  `timescale 1 ns / 100 ps
2
3  module reg_alu (
4      input wire clk, reset, sel, wr,
5      input wire [1:0] op,
6      input wire [2:0] rd_addr_a, rd_addr_b, wr_addr,
7      input wire [15:0] d_in,
8      output wire [15:0] d_out_a, d_out_b,
9      output wire cout
10 );
11
12     reg [15:0] reg_file [7:0];
13     wire [15:0] alu_out;
14     wire [15:0] reg_a, reg_b;
15     wire alu_cout;
16
17     assign reg_a = reg_file[rd_addr_a];
18     assign reg_b = reg_file[rd_addr_b];
19     assign d_out_a = (sel) ? alu_out : reg_a;
20     assign d_out_b = reg_b;
21
22     always @(posedge clk or posedge reset) begin
23         if (reset) begin
24             // Reset all registers to 0
25             reg_file[0] <= 16'h0000;
26             reg_file[1] <= 16'h0000;
27             reg_file[2] <= 16'h0000;
28             reg_file[3] <= 16'h0000;
29             reg_file[4] <= 16'h0000;
30             reg_file[5] <= 16'h0000;
31             reg_file[6] <= 16'h0000;
32             reg_file[7] <= 16'h0000;
33         end else if (wr) begin
34             reg_file[wr_addr] <= d_in;
35         end
36     end
37
38     alu alu_inst (
39         .op(op),                // ALU operation
40         .i0(reg_a),             // First input (from reg_a)
41         .i1(reg_b),             // Second input (from reg_b)
42         .o(alu_out),            // ALU output
43         .cout(alu_cout)         // Carry out
44     );
45
46     assign cout = alu_cout;
47
48 endmodule
```

tb_reg_alu.v:-

≡ reg_alu.v U

≡ tb_reg_alu.v U X

week6 > ≡ tb_reg_alu.v

```
1  | timescale 1 ns / 100 ps
2  | `define TESTVECS 8
3
4  module tb;
5      reg clk, reset, wr, sel;
6      reg [1:0] op;
7      reg [2:0] rd_addr_a, rd_addr_b, wr_addr; reg [15:0] d_in;
8      wire [15:0] d_out_a, d_out_b;
9      reg [28:0] test_vecs [0:(`TESTVECS-1)];
10     integer i;
11     initial begin $dumpfile("tb_reg_alu.vcd"); $dumpvars(0,tb); end
12     initial begin reset = 1'b1; #12.5 reset = 1'b0; end
13     initial clk = 1'b0; always #5 clk =~ clk;
14     initial begin
15         test_vecs[0][28] = 1'b0; test_vecs[0][27] = 1'b1; test_vecs[0][26:25] = 2'bxx;
16         test_vecs[0][24:22] = 3'ox; test_vecs[0][21:19] = 3'ox;
17         test_vecs[0][18:16] = 3'h3; test_vecs[0][15:0] = 16'hcdef;
18
19         test_vecs[1][28] = 1'b0; test_vecs[1][27] = 1'b1; test_vecs[1][26:25] = 2'bxx;
20         test_vecs[1][24:22] = 3'ox; test_vecs[1][21:19] = 3'ox;
21         test_vecs[1][18:16] = 3'o7; test_vecs[1][15:0] = 16'h3210;
22
23         test_vecs[2][28] = 1'b0; test_vecs[2][27] = 1'b1; test_vecs[2][26:25] = 2'bxx;
24         test_vecs[2][24:22] = 3'o3; test_vecs[2][21:19] = 3'o7;
25         test_vecs[2][18:16] = 3'o5; test_vecs[2][15:0] = 16'h4567;
26
27         test_vecs[3][28] = 1'b0; test_vecs[3][27] = 1'b1; test_vecs[3][26:25] = 2'bxx;
28         test_vecs[3][24:22] = 3'o1; test_vecs[3][21:19] = 3'o5;
29         test_vecs[3][18:16] = 3'o1; test_vecs[3][15:0] = 16'hba98;
30
31         test_vecs[4][28] = 1'b0; test_vecs[4][27] = 1'b0; test_vecs[4][26:25] = 2'bxx;
32         test_vecs[4][24:22] = 3'o1; test_vecs[4][21:19] = 3'o5;
33         test_vecs[4][18:16] = 3'o1; test_vecs[4][15:0] = 16'hxxxx;
34
35         test_vecs[5][28] = 1'b1; test_vecs[5][27] = 1'b1; test_vecs[5][26:25] = 2'b00;
36         test_vecs[5][24:22] = 3'o1; test_vecs[5][21:19] = 3'o5;
37         test_vecs[5][18:16] = 3'o2; test_vecs[5][15:0] = 16'hxxxx;
38
39         test_vecs[6][28] = 1'b1; test_vecs[6][27] = 1'b1; test_vecs[6][26:25] = 2'b01;
40         test_vecs[6][24:22] = 3'o2; test_vecs[6][21:19] = 3'o7;
41         test_vecs[6][18:16] = 3'o4; test_vecs[6][15:0] = 16'hxxxx;
42
43         test_vecs[7][28] = 1'b1; test_vecs[7][27] = 1'b0; test_vecs[7][26:25] = 2'b01;
44         test_vecs[7][24:22] = 3'o4; test_vecs[7][21:19] = 3'o4;
45         test_vecs[7][18:16] = 3'ox; test_vecs[7][15:0] = 16'hxxxx;
46     end
47     initial {sel, wr, op, rd_addr_a, rd_addr_b, wr_addr, d_in} = 0;
48     reg_alu reg_alu_0 (clk, reset, sel, wr, op, rd_addr_a, rd_addr_b, wr_addr, d_in,
49         d_out_a, d_out_b, cout);
50     initial begin
51         #6 for(i=0;i<`TESTVECS;i=i+1)
52             begin #10 {sel, wr, op, rd_addr_a, rd_addr_b, wr_addr, d_in}=test_vecs[i]; end
53         #100 $finish;
54     end
```

Verilog VVP Output Screenshot:-

VCD info: dumpfile tb_reg_alu.vcd opened for output.

Time	sel	wr	op	rd_a	rd_b	wr_addr	d_in	d_out_a	d_out_b	cout
200	0	1	xx	xxx	xxx	011	cdef	0000	0000	x
340	0	1	xx	xxx	xxx	111	3210	XXXx	XXXx	x
480	0	1	xx	011	111	101	4567	0000	3210	x
620	0	1	xx	001	101	001	ba98	0000	4567	x
760	0	0	xx	001	101	001	xxxx	0000	4567	x
900	1	1	00	001	101	010	xxxx	0000	4567	0
1040	1	1	01	010	111	100	xxxx	4567	3210	1
1180	1	0	01	100	100	xxx	xxxx	ba98	ba98	1

GTKWAVE Output(tb_reg_alu.vcd):-

